Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. It focuses on developing computer programs that can access data and use it to learn for themselves.

Supervised Learning

Supervised learning is the most common paradigm in machine learning. In this approach, models are trained on labeled data, where each input is paired with the correct output. Common algorithms include linear regression, logistic regression, decision trees, and neural networks.

Classification

Classification is a supervised learning task where the output is a discrete category. Examples include spam detection, image recognition, and medical diagnosis. The model learns to map inputs to one of several predefined classes.

Regression

Regression is another supervised learning task, but the output is a continuous value. Examples

include predicting house prices, stock prices, or temperature. The model learns a function that maps inputs to continuous outputs.

Unsupervised Learning

Unsupervised learning deals with unlabeled data. The model tries to find hidden patterns or intrinsic structures in the input data. Common techniques include clustering, dimensionality reduction, and anomaly detection.

Reinforcement Learning

Reinforcement learning is a paradigm where an agent learns to make decisions by interacting with an environment. The agent receives rewards or penalties for its actions and learns to maximize the cumulative reward over time. This approach has been successfully applied to game playing, robotics, and autonomous driving.